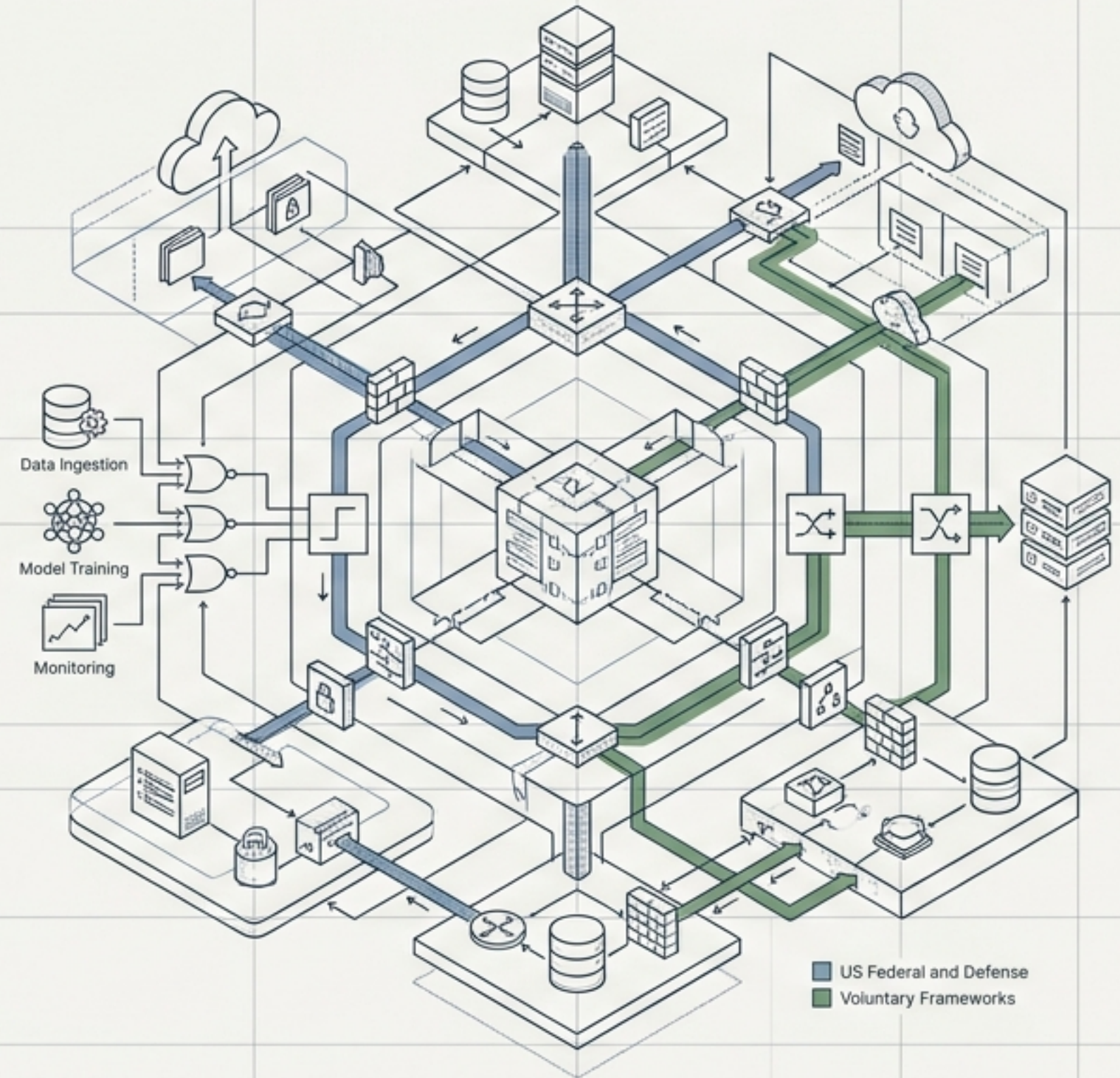


Week 5 — Regulation and Responsible AI

Transforming legal and regulatory frameworks into pipeline requirements and system constraints.

[Level: 100-300]

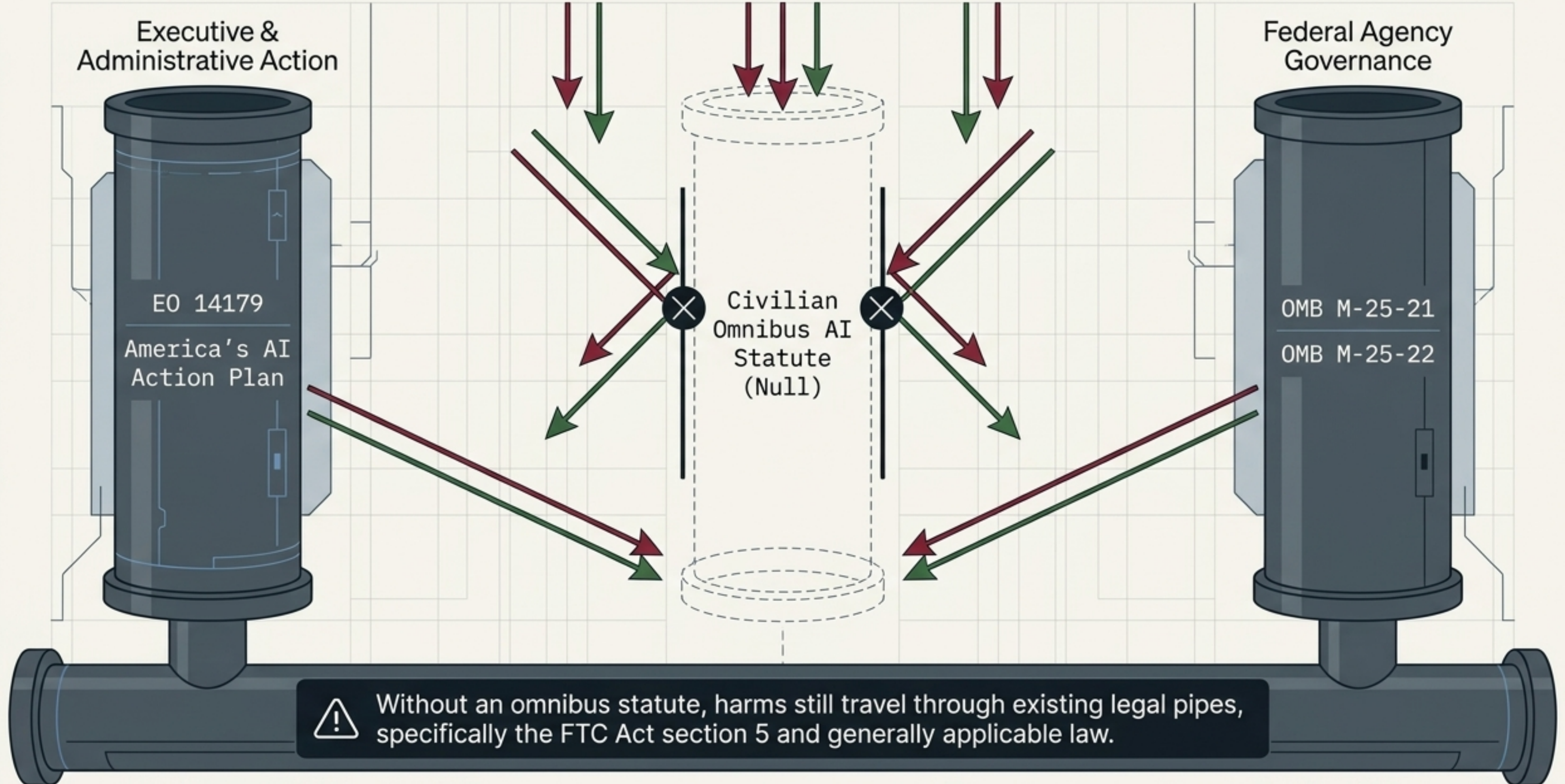


The Architect's Disclaimer



```
// Execution Context: Educational Purpose Only.  
// Status: Not_Legal_Advice.  
// Objective: Mapping pipeline requirements, not  
practicing law.  
// Directive: Do not collapse disparate frameworks  
into a single checklist.
```

US Civilian Posture operates through existing pipelines



Defense mandates establish distinct perimeters and futures

Exploratory Sandbox

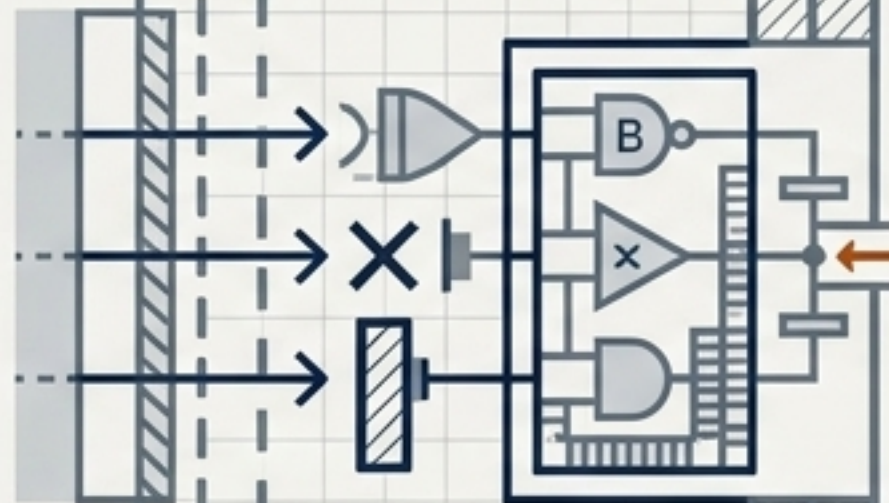
FY2026 NDAA

Sections 1532–1535:

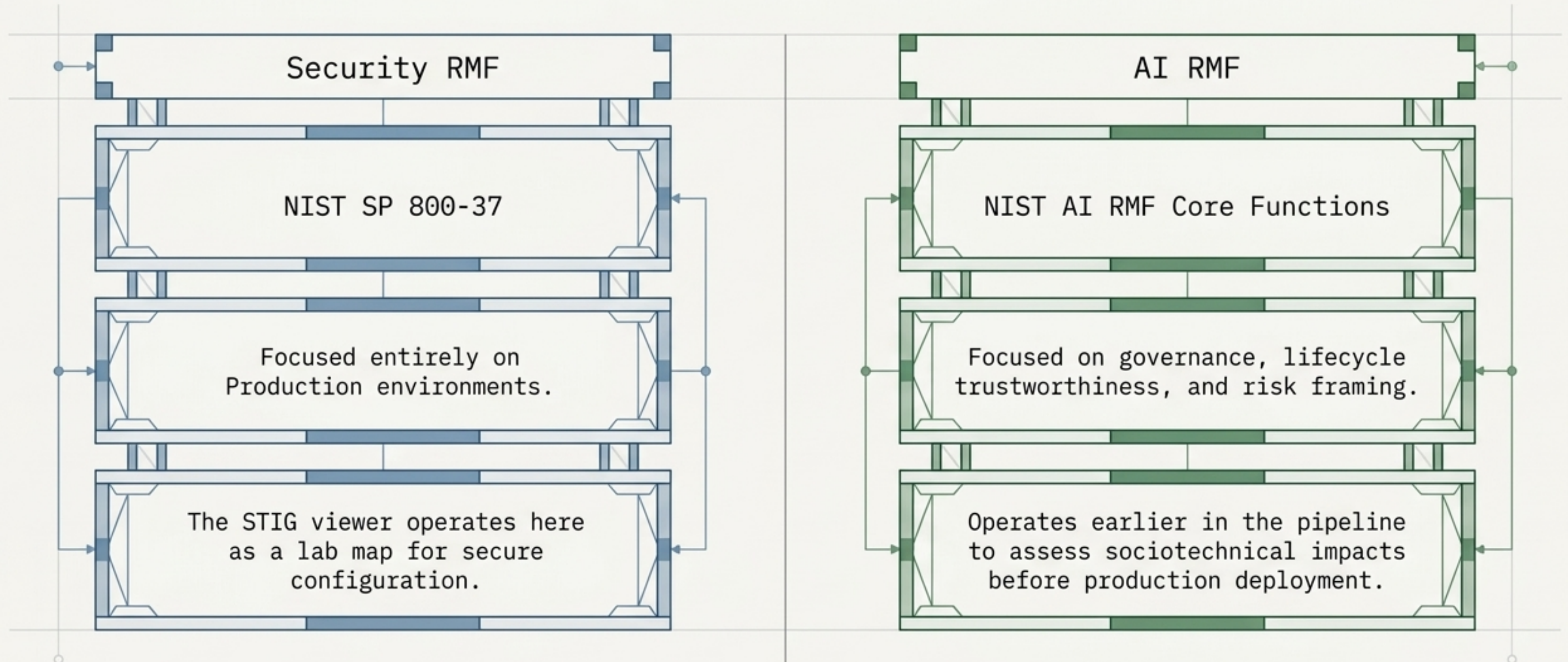
- Covered-AI screening
- Assessment protocols
- Authorized Sandboxes
- Futures Committee

Production Environment

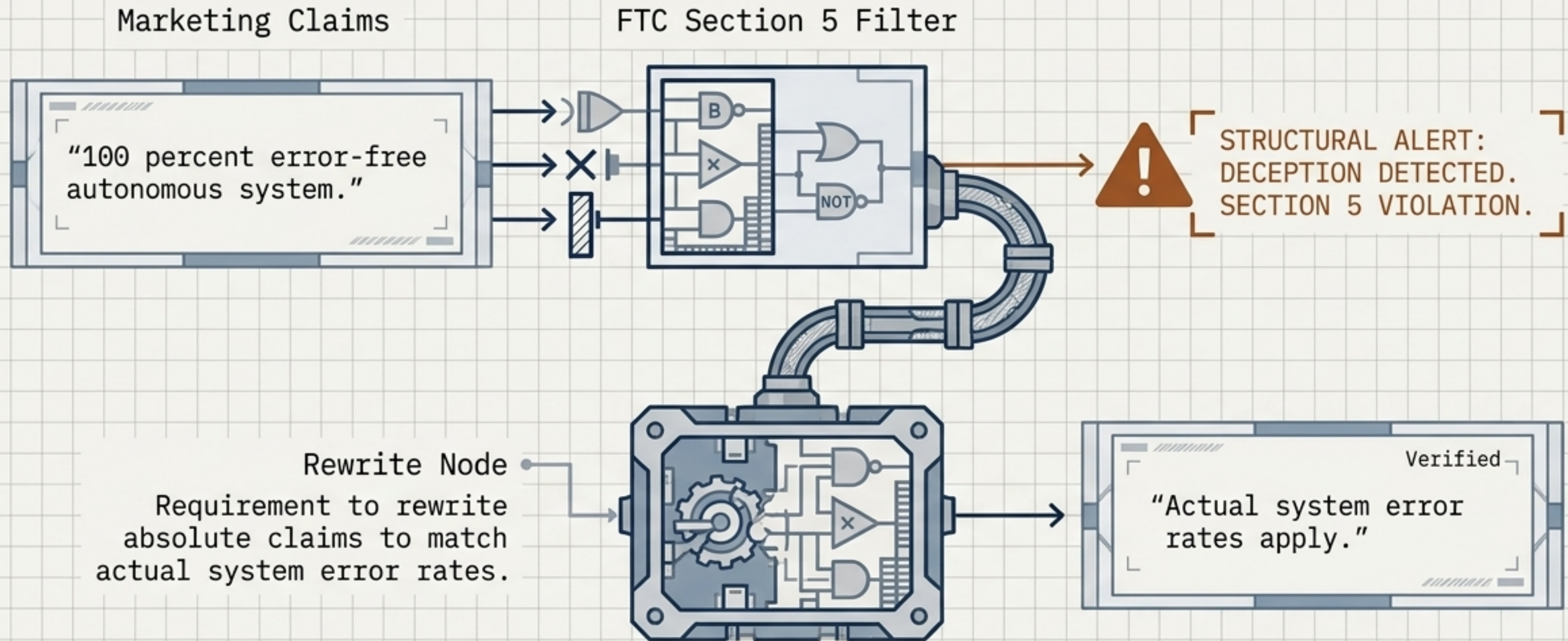
⚠ A sandbox is strictly exploratory. It is not an Authorization to Operate (ATO).



Distinct frameworks for distinct deployment phases

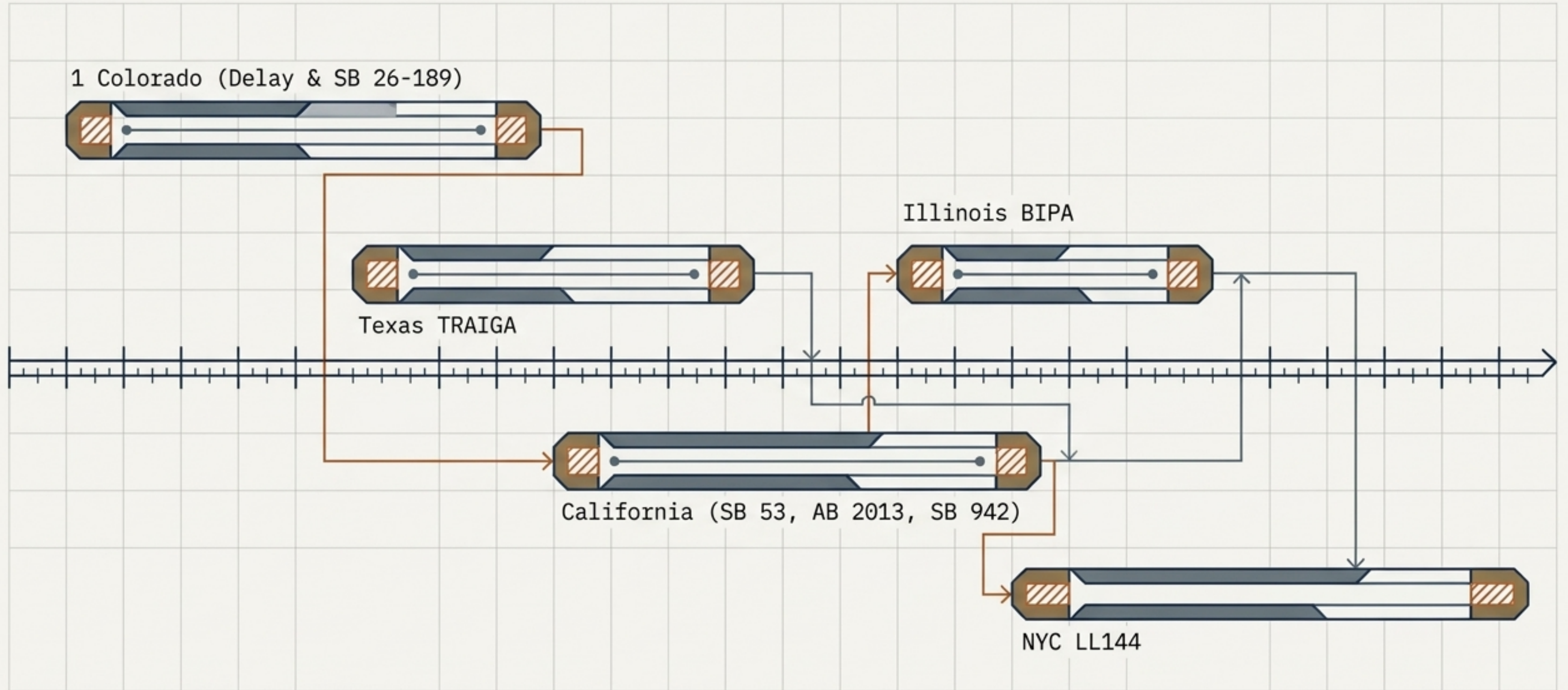


FTC enforcement targets deception and unfairness



Operation AI Comply is an enforcement posture signaling increased scrutiny, not a new congressional statute.

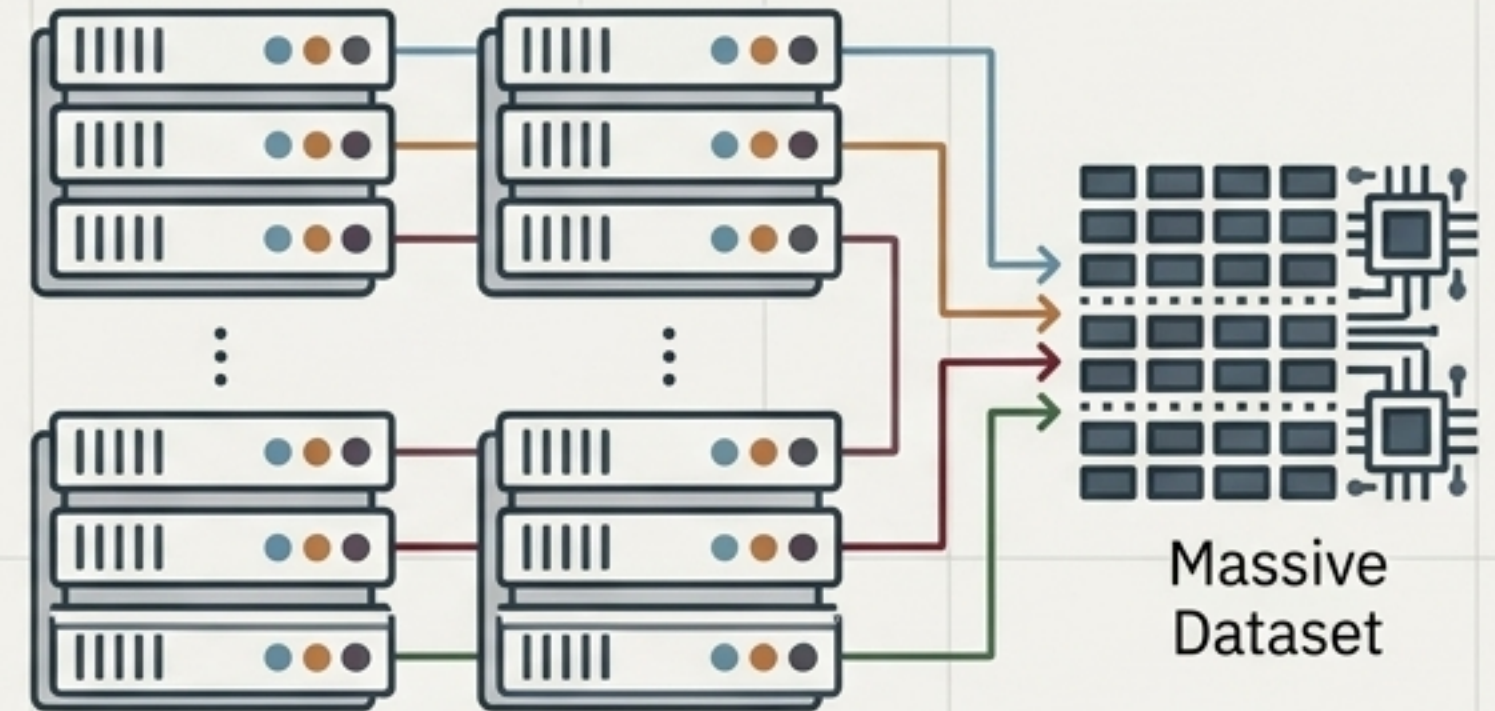
The state-level deployment timeline



Distinguishing high-risk product duties from frontier-lab duties



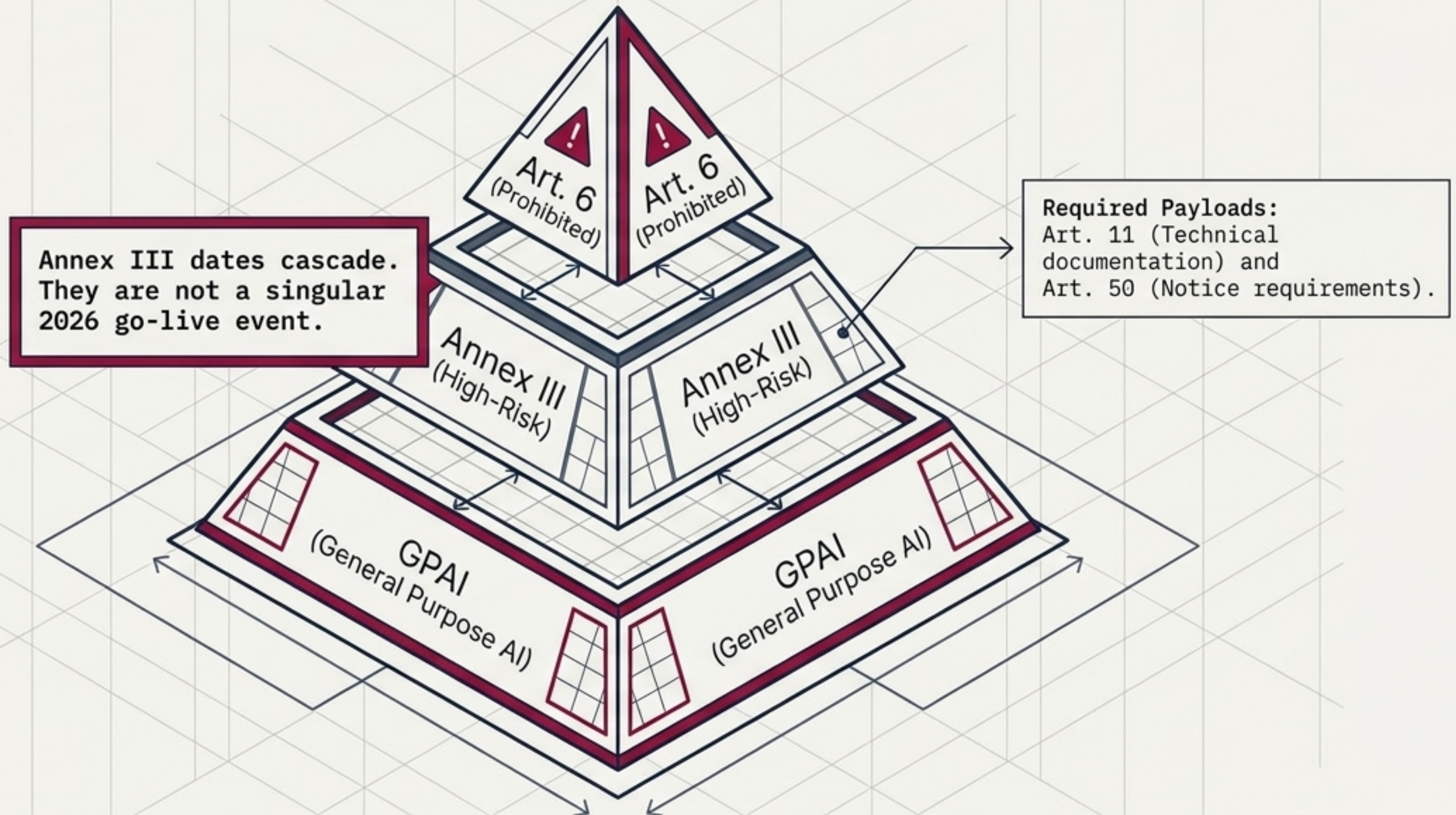
High-risk product duties apply to downstream deployment and specific end-user use cases.



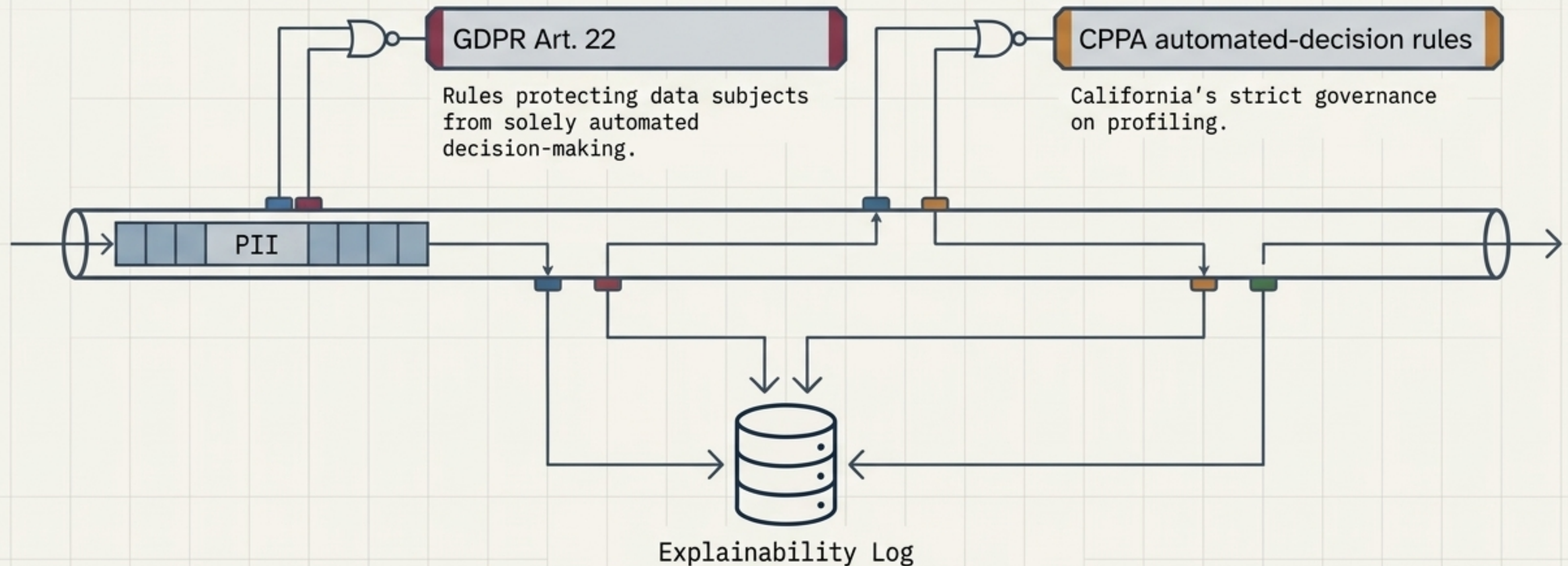
Frontier-lab duties apply to massive computational thresholds and foundational model training

A bank using a vendor model **is not a large frontier developer** under the New York RAISE Act. **Know your exact node in the supply chain.**

Mapping the EU AI Act risk topology

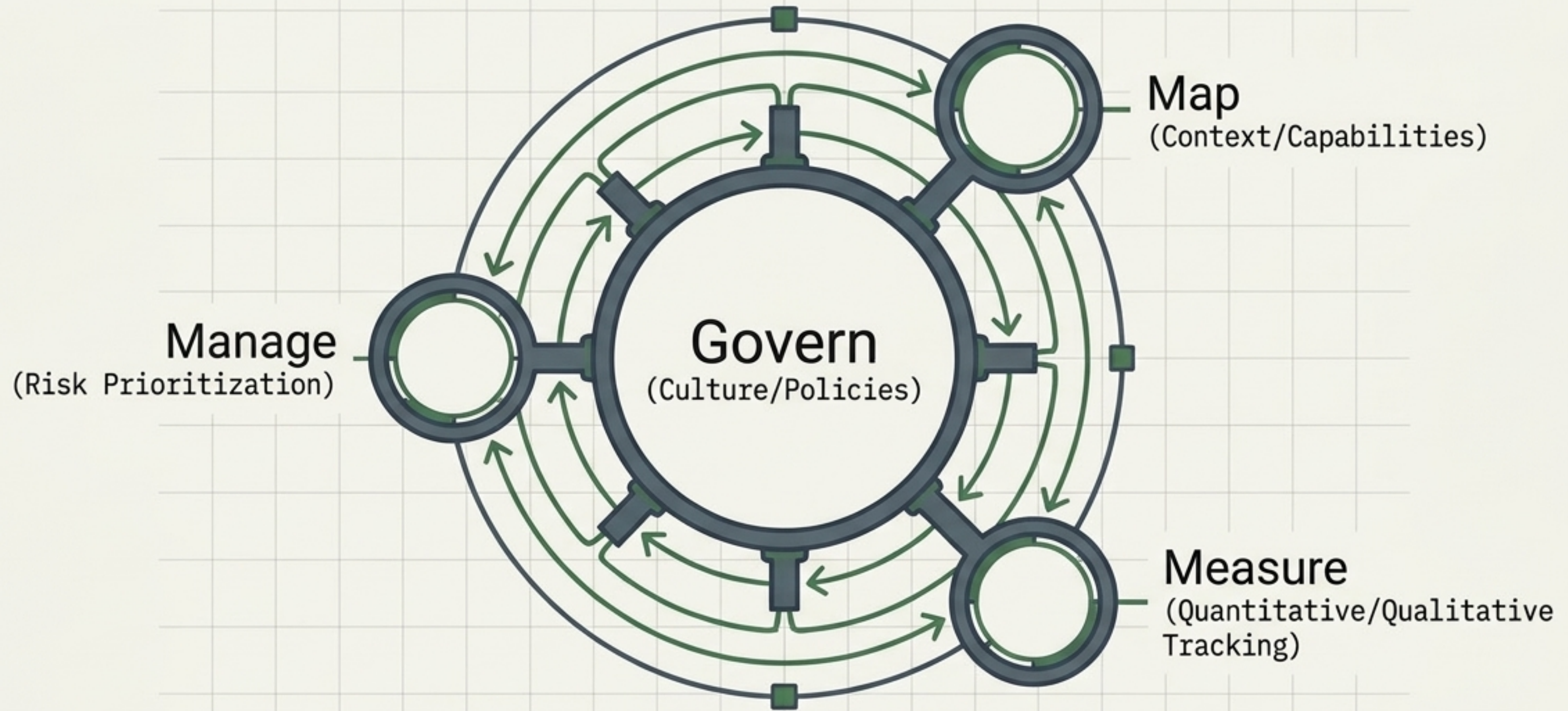


Intersecting AI pipelines with established privacy constraints



Both frameworks necessitate robust, immutable explainability logs baked directly into the system architecture.

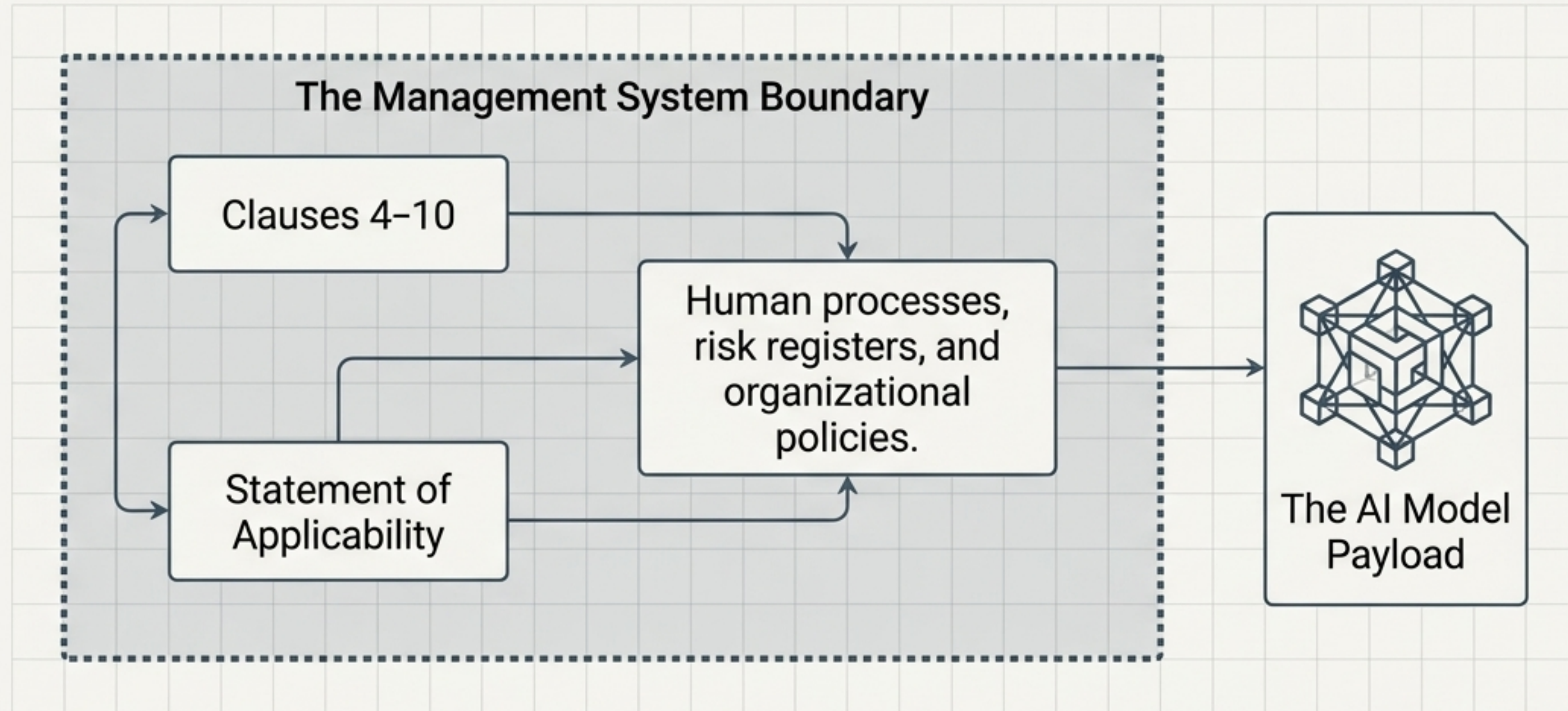
NIST AI RMF core architectural functions



Precision mapping: NIST Measure versus TEVV

NIST Measure	TEVV
Measure is the formal Function within the NIST AI RMF.	Test, Evaluation, Verification, and Validation.
Specifically, it maps to Table 3 in the framework. Crucial detail: The assessment of Fairness resides precisely in Measure 2.11.	TEVV is the methodology used to execute the measurement. TEVV is not a fifth function of the NIST AI RMF.

Defining the ISO/IEC 42001 boundary



The ISO/IEC 42001 certificate covers the management system—the human processes, documentation, and governance surrounding the AI—it explicitly does not certify the safety of the model itself.

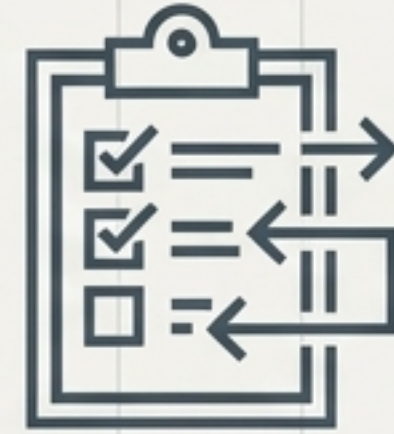
Separating the blueprint from the audit



The Blueprint

ISO/IEC 42001 Clauses

Specifically Clauses 4-10, which define the requirements the enterprise architect must build into the management system.



The Inspection

Stage 1 and Stage 2

These are strictly ISO/IEC 17021-1 audit steps executed by external certifiers, they are not clauses within the 42001 standard.

Lexicon disambiguation for system architects

> TYPE: Words

> SOURCE: ISO/IEC 22989

> PAYLOAD: Establishes the vocabulary and core concepts.

> TYPE: Methods

> SOURCE: ISO/IEC 23894 and AI RMF

> PAYLOAD: Provides the risk management guidance and methodologies.

> TYPE: Obligations

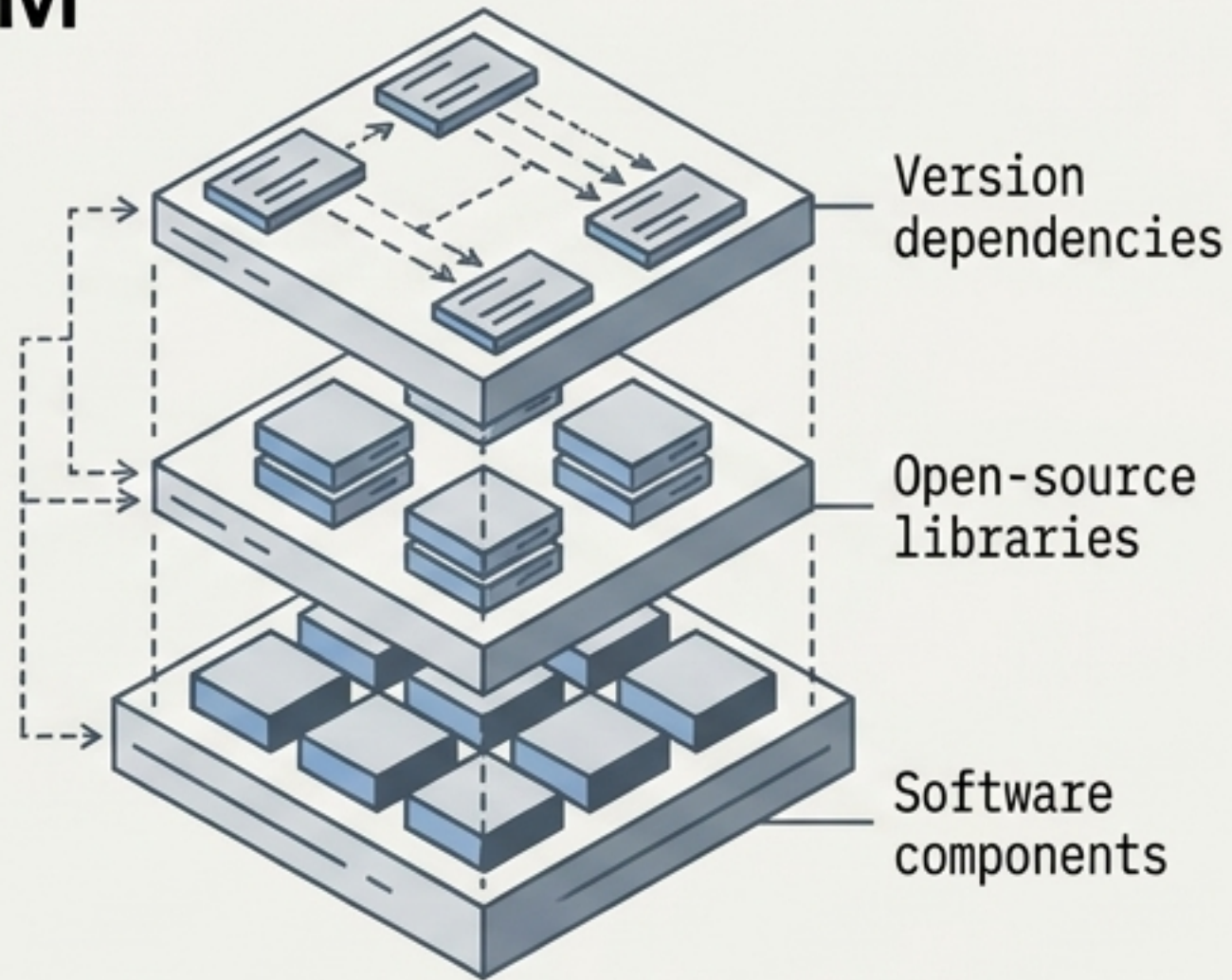
> SOURCE: ISO/IEC 42001 and specific statutes

> PAYLOAD: Dictates the enforceable requirements.

WARNING: NIST AI 600-1 is a generative AI profile detailing specific use-case risks; it is not a statute.

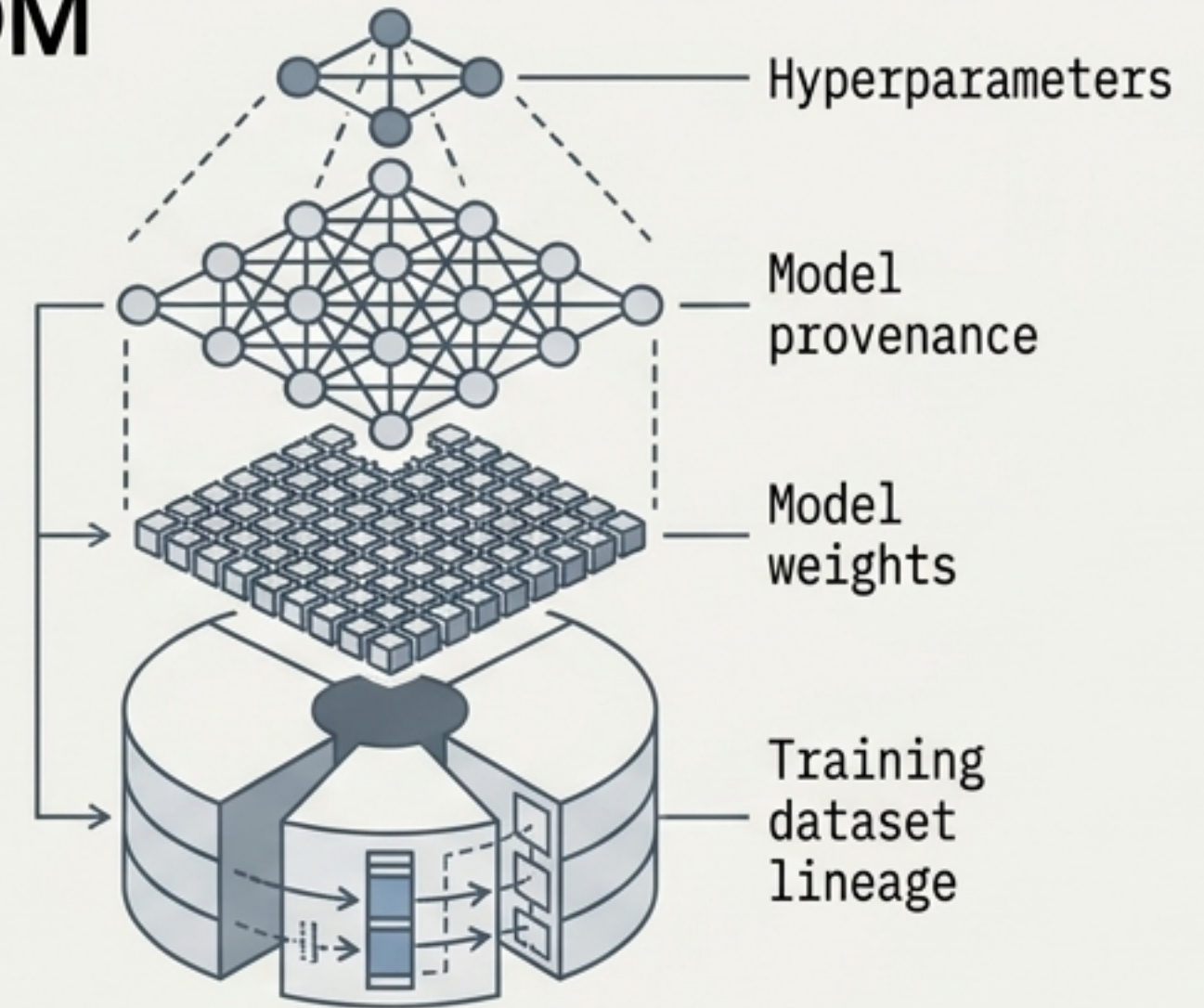
Artifact matrices: SBOM versus AIBOM

SBOM



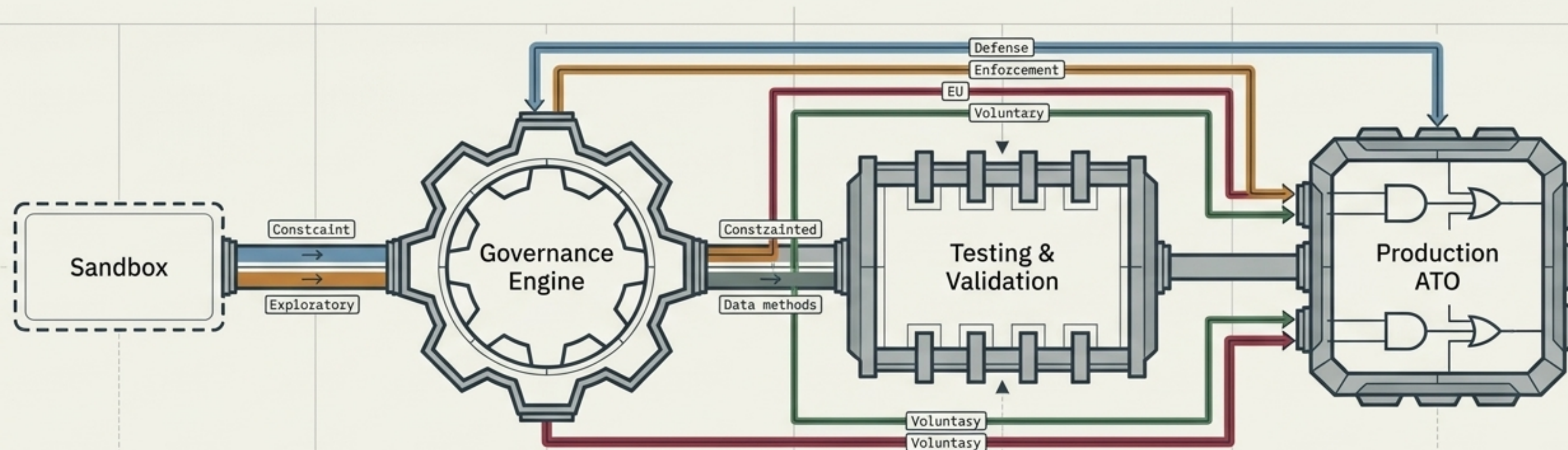
Software Bill of Materials. Maps traditional software components and dependencies.

AIBOM



AI Bill of Materials. Maps the model's distinct mathematical and data lineage.

The AI Architecture Control Plane



One-Page Applicable Law & Framework Summary

US Civilian	E0 14179	America's AI Action Plan	OMB M-25-21/22	FTC Sec 5	
US Defense	FY2026 NDAA (1532-1535)				
State Level	CO (SB 26-189)	TX TRAIGA	CA (SB 53, AB 2013, SB 942)	IL BIPA	NYC LL144
European Union	EU AI Act (Art 6, Art 11, Art 50, Annex III)				
Voluntary Standards	NIST AI RMF	ISO/IEC 42001	Security RMF (SP 800-37)		